

MONOIDS FOR COMPUTERS

LUCÍA ASENSIO MARTÍN, NEWCASTLE UNIVERSITY

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②. Question $\rightsquigarrow$ Find (short, fast) normal forms (2015)

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② Question $\rightsquigarrow$ Find (short, fast) normal forms (2015)

(J. Britnell, A. Duncan, D. Francoeur, S. Rees at Ncl)

①. What are step trace monoids?

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Trace monoids ----- independence ----- determines presentation
(partially commutative graph
monoids, RAAMs)

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+ concurrency

Step trace monoids ----- indep/concurrency ----- determine presentation

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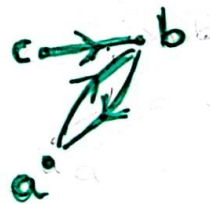
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Step trace monoids ----- indep/concurrency graphs ----- determine presentation



$$M = \langle a, b, c, \{a, c\}, \{b, c\} \mid ab=ba, \{b, c\} = cb \rangle$$

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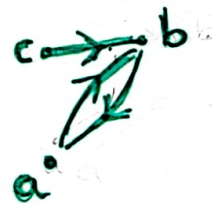
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$$M = \langle a, b, c, \{a, c\}, \{b, c\} \mid \underbrace{ab=ba}_{AB=BA}, \underbrace{\{b, c\}=cb}_{ALB=A \cdot B} \rangle$$

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A word in the generators $\longrightarrow$ an execution of a program

Equivalent words $\longrightarrow$ equivalent executions of program

Shortest word for element $\longrightarrow$ most efficient way to run program

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Shortest normal form for step trace monoids?

②. A question : short normal forms?

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Unique graph
↳ vertices for actions
↳ 3 types of edges
describe dependence



element of
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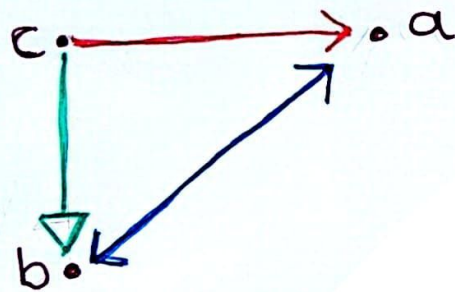
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 $a \xrightarrow{\text{red}} b \quad a < b$



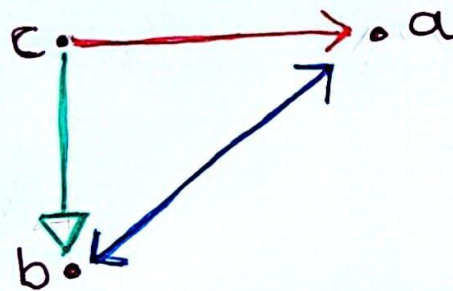
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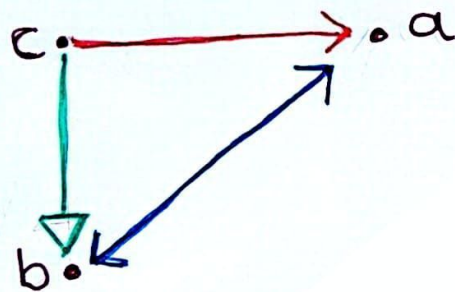
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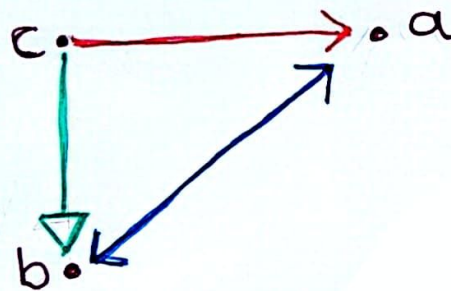
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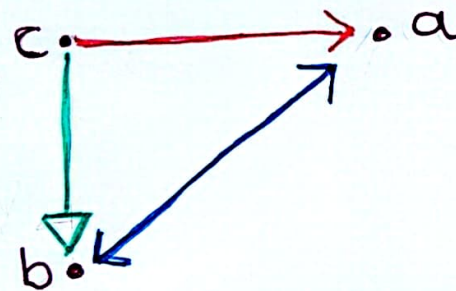
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- "Rearranging" graph gives all words representing same element.
- The arrows tell us allowed rearrangements.

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- Only $\rightarrow, \rightarrow$: ✓

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- Only $\rightarrow$ edges : $\checkmark$ (Cartier-Foata for trace monoids)

- Only $\rightarrow$, $\rightarrow$: $\checkmark$

- $\rightarrow$, $\rightarrow$, $\leftrightarrow$: ?

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So far: elements whose graphs have...

- Only $\rightarrow$ edges : ✓ (Cartier-Foata for trace monoids)

- Only $\rightarrow$, $\rightarrow$: ✓

- $\rightarrow$, $\rightarrow$, $\leftrightarrow$: ?

Slow, shortest normal forms ✓

Fast, long normal forms ✓

Fast and short?

② A question: short normal forms?

So far: elements whose graphs have...

- Only $\rightarrow$ edges : ✓ (Cartier-Foata for trace monoids)

- Only $\rightarrow$, $\rightarrow$: ✓

- $\rightarrow$, $\rightarrow$, $\leftrightarrow$: ?

Slow, shortest normal forms ✓

Fast, long normal forms ✓

Fast and short?

Thanks :)